Graphic Era HILL UNIVERSITY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAJOR PROJECT PROPOSAL

FINAL YEAR

2025-2026

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Project Proposal: Smart AI-Powered Door Lock System

Objective:

Build a secure smart door lock that combines motion detection, AI human/non-human classification, and face-based access control with admin override, plus a React dashboard for monitoring and approvals.

Key Features:

- **Motion Detection (Continuous Surveillance)**
  1. *Camera runs an edge motion filter (frame-differencing / OpenCV MOG2) to detect major movement.*
  2. *Debounce + adaptive thresholding to avoid noise (light flicker, small pets).*
  3. *When motion score crosses threshold → trigger object detection.*
  4. *Only major events (high motion + detection confidence) are sent to the server and saved (thumbnail, timestamps, deviceId).*

- **Human vs Non-Human Classification**
  1. *Lightweight detector (YOLO/MobileNet) classifies detections as human or non_human.*
  2. *Human events are persisted in MongoDB and shown in Detection History on the dashboard.*

- **Face Recognition Access (Button Mode)**
  1. *Button press captures short burst of frames → face detect → embedding (FaceNet/ArcFace).*
  2. *If match with saved user (embedding similarity ≥ threshold) → unlock automatically.*
  3. *If no confident match → send approval request to admin via real-time notification; admin can Approve/Reject.*

- **React Dashboard (User/Admin)**
  1. *Profile, Detection History (filters & thumbnails), Saved Users (Face Library), Access Requests & Approvals, Device Health, Reports.*
  2. *Real-time via Socket.IO; JWT + role-based access.*

- **Backend & Data**
  1. *Node.js/Express API, Socket.IO, MongoDB (embeddings + optional GridFS for images).*
  2. *Collections: users, faces, detections, access_logs, devices, approvals.*
- **Hardware & Edge**

1. *Raspberry Pi / Jetson Nano runs motion + detection + face pipeline; controls lock via GPIO; button input.*
2. *Edge inference for low latency; server for storage, approvals, and analytics.*

- **Security & Privacy**
  - *JWT auth, RBAC, TLS, encrypt embeddings at rest, retention policies, rate limiting.*

Outcome

A robust smart lock that reduces false alerts with motion gating, provides reliable face-based unlocking, and preserves an admin-in-the-loop safety net — all visible and controllable through a secure React dashboard.